# Annex 2: CAN Parameterization

# Instruction Manual

Manual Version: V2.04

Update Date: 2023.07.01

CONTENTS

# I. Overview

In order to facilitate the needs of different users, the USB-CAN adapter provides two modes for parameters setting:

1. Simple mode, which provides simple options for baud rate, operating mode, filter mode, etc. When using the USB-CAN Tool debugging tool or user-defined programming, you only need to make a simple selection of settings to quickly configure the adapter, which is suitable for users who are beginners or who have just come into contact with CAN communication;

2. Professional mode, in which users can customize the baud rate and hardware filtering mode by modifying various register parameters, suitable for users who are more familiar with CAN communication or enthusiasts.

These two modes are free to choose according to your use.

**Caution:**

**When the CAN bus is sending and receiving data normally, try not to modify the CAN bus parameters or turn off the CAN bus through the USBCAN adapter, you should wait until the data sending and receiving stops or disconnect the USBCAN adapter from the CAN bus before carrying out the corresponding operation.**

# II.Baud Rate Setting

## 2.1 SJA1000 Compatible Mode

To facilitate (SJA1000) CAN controller users, USB-CAN adapter to do the corresponding compatibility processing, users only need to configure the corresponding Timing0 (BTR0), Timing1 (BTR1) register value, you can configure the corresponding baud rate. And you do not have to touch the relatively complex, specialized multiple register parameters.

Conventional baud rate index value comparison table is as follows:

| CAN Baud | Timing0(BTR0) | Timing1(BTR1) |
|---|---|---|
| 10 Kbps | 0x31 | 0x1C |
| 20 Kbps | 0x18 | 0x1C |
| 40 Kbps | 0x87 | 0xFF |
| 50 Kbps | 0x09 | 0x1C |
| 80 Kbps | 0x83 | 0xFF |
| 100 Kbps | 0x04 | 0x1C |
| 125 Kbps | 0x03 | 0x1C |
| 200 Kbps | 0x81 | 0xFA |
| 250 Kbps | 0x01 | 0x1C |
| 400 Kbps | 0x80 | 0xFA |
| 500 Kbps | 0x00 | 0x1C |
| 666 Kbps | 0x80 | 0xB6 |
| 800 Kbps | 0x00 | 0x16 |
| 1000 Kbps | 0x00 | 0x14 |
| 33.33 Kbps | 0x09 | 0x6F |
| 66.66 Kbps | 0x04 | 0x6F |
| 83.33 Kbps | 0x03 | 0x6F |

**Note:**

**1. When configuring the baud rate, users only need to follow the baud rate**

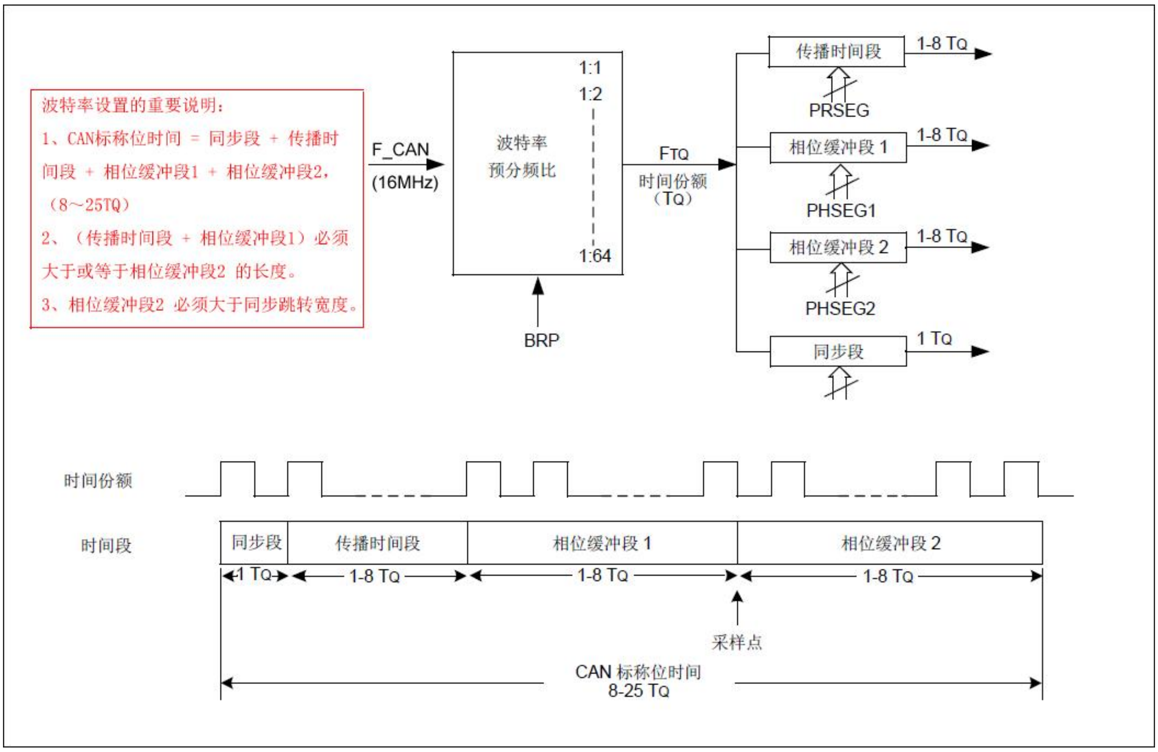
**parameters given by SJA1000 (16MHz) to set up.**

**2. Conventional baud rate can be configured according to the value in the above table. For other unconventional baud rates, you can use the included baud rate detection tool to detect and get the corresponding baud rate parameters. Maybe you can also use the baud rate calculator in the USB_CAN TOOL installation directory to calculate the baud rate. (Refer to *"6. Plug-in 2: Baud Rate Detection Tool User Manual.pdf"*)**

**3. This adapter does not support baud rate below 10K for the time being.**

## 2.2 Advanced Mode

In the advanced mode of baud rate setting, the user changes the desired baud rate by setting the CAN register values. There are seven registers related to the baud rate: Synchronization Jump Width, Preshunt, Sample Point, Phase Buffer Segment 2 Select Bit, Synchronization Segment, Propagation Time Segment, Phase Buffer Segment 1, and Phase Buffer Segment 2. The following bit timing diagram illustrates in detail the relationship between these registers:



**A few important notes on CAN custom baud rate settings:**

*1), CAN Nominal Bit Time = Synchronization Segment + Propagation Time Segment + Phase Buffer Segment 1 + Phase Buffer Segment 2, (8 to 25 TQ).*

*2), (Propagation time segment + Phase buffer segment 1) must be greater than or equal to the length of phase buffer segment 2.*

*3), Phase buffer segment 2 must be greater than the synchronization hop width.*

Baud rate calculation formula = 16000000/(synchronization segment + propagation time segment + phase buffer segment 1 + phase buffer segment 2)/pre-divided frequency.

*The theoretical value range of each register is shown in the table below:*

| Abbreviation | Description | Range of Values | Note |
|---|---|---|---|
| SJW | Synchronized Jump Width | 1~4 | |
| BRP | Pre - Crossover | 1~64 | |
| SAM | Sampling Point | 0~1 | *0*- One - Time Sampling<br>*1*- Three - Time Sampling |
| SYN | Synchronization Segment | 1 | This Value Is Forced To 1 and Does Not Need to Be Set By The User |
| PHSEG2_SEL | Phase Buffer Segment 2 Selection Bit | 0-1 | *0*-Determined by phase buffer segment 1 time, when the value of phase buffer segment 2 is forced to be equal to phase buffer segment 1<br>*1*-Programmable |
| PRSEG | Dissemination Time Period | 1~8 | |
| PHSEG1 | Phase Buffer Segment 1 | 1~8 | |
| PHSEG2 | Phase Buffer Segment 2 | 1~8 | |

# III.Working Modalities

The USB-CAN adapter supports three operating modes: normal operation, listen-only mode, and self-test mode (loopback mode). The value comparison table for these three operating modes is shown below:

| Operating Mode | Retrieve Value | Note |
|---|---|---|
| Normal Operation | 0 | The CAN module appears on the CAN bus and can send and receive CAN messages. |
| Listen-Only Mode | 1 | If the Listen Only mode is activated, the module will be present on the CAN bus, but in a passive state. It will receive telegrams, but ti will not send them or answer signals. This mode can be used as a bus monitor because the CAN bus does not affect the data communication. |
| Self-Test Mode (Loopback Mode) | 2 | The self-test (loopback) mode is used for the adapter to perform a self-test, allowing the CAN module to receive its own messages. In this mode, the CAN module's transmit path is internally connected to the receive path. In this mode, a "false" answer is provided so that another node is not required to provide the answer bit and the CAN message is not actually sent to the CAN bus. CAN messages are not actually sent to the CAN bus. CAN frames sent by the adapter will be received back by the adapter. |

# IV.Filter Settings

## 4.1 Filter Registers

The filter register consists of two 32-bit registers: the filter acceptance filter (ACR) and the filter mask register (AMR). The filtering function for CAN received messages is realized by setting the values of these two registers through software.

When the CAN module receives a message, it compares the message identifier with the corresponding bit in the filter. If the identifier matches the user-configured filter, the message is stored in the corresponding receive cache queue of the CAN controller.

The receive mask can be used to ignore the select bits of the identifier during reception. These bits will not be compared to the bits in the filter when the message is received. For example, if the user wishes to receive all messages with identifiers 0, 1, 2, and 3, the user would need to mask out the lower 2 bits of the identifier. A bit of the Masking Register equal to 1 indicates that the filtering of the bit corresponding to the ID bit is ignored, e.g., if the Masking Register value = FFFFFFFFFF, then all messages can be received.

The acceptance filter ACR, and the acceptance mask AMR are both 32bits (4bytes). For ID values that require acceptance filtering, the highest bit of the ID (Bit10 for the highest bit of the ID in standard frames, Bit28 for extended frames) is aligned with the highest bit (Bit31) of the ACR/AMR, i.e., left-aligned.

Both the CAN bus acceptance filter and the acceptance mask are for CAN reception. Note: When AMR is FF FF FF FF FF, it means that all filter bits of ACR are masked, i.e. all messages can be received.

Note: For a detailed description of the ID format, please refer to: *"8. Annex 1: ID Alignment.pdf"* instruction document.

Examples:

(1) Standard frame example: To receive standard frames, the filtering method needs to be selected as "Receive all types" or "Receive standard frames only", ACR=Any value, AMR= 0xFFFFFFFF, the adapter can receive CAN messages with

any ID; ACR=0x6F000000, AMR=0x00FFFFFF, the adapter can receive frames with ID 0x378 to 0x37F.

(2) Extended frame example: To receive extended frames, the filtering method needs to be selected as "Receive all types" or "Receive extended frames only", ACR=Any value, AMR= 0xFFFFFFFF, the adapter can receive CAN messages with any ID; ACR=0x00001BC0, AMR=0x0000003F, the adapter can receive CAN messages with ID=0x00000378 to 0x0000037F.

## 4.2 Filter modes

The USB-CAN adapter provides three filter modes: receive all types, receive only standard frames, and receive only extended frames. The following table shows a comparison of the filter mode values:

| Value | Title | Instruction |
|---|---|---|
| 1 | Receive All Types | The filter allows reception of both standard and extended frames. |
| 2 | Receive Only Standard Frames | The filter allows standard frames to be received; extended frames will not pass. |
| 3 | Receive Extended Frames Only | The filter allows reception of extended frames; standard frames will not pass. |

Note: For a more detailed description of the filter settings, please refer to the *"Annex 4: CAN Filter Settings"* document.